

ARTIFICIAL INTELLIGENCE, ALGORITHMIC RECOMMENDATIONS AND COMPETITION

EMILIO CALVANO

Università LUISS (Rome), Toulouse School of Economics and CEPR

GIACOMO CALZOLARI

European University Institute, Bologna University and CEPR

VINCENZO DENICOLÒ

Bologna University and CEPR

SERGIO PASTORELLO

Bologna University

We study how algorithmic recommendations affect product-market competition. We embed recommender systems, trained on sparse and noisy feedback, in a model of consumer search and price competition. Recommendations create personalized prominence: consumers start searching from products predicted to match their tastes. This improves matching and reduces search costs, but also raises equilibrium prices. Prices increase through front-loading of high-value consumers, reliance on informative recommendations, and reduced price salience under directed search. Due to the higher prices, consumer surplus may fall despite better matches. We also show that manipulated recommendations can intensify competition and limit the profitability of self-preferencing.

1. INTRODUCTION

To help consumers navigate the vast array of products available online, digital platforms increasingly rely on *recommender systems* (RSs): AI-based algorithms that use feedback such as ratings, clicks, or other forms of user activity to predict consumers' interest in products they have not yet experienced. Platforms use these predictions to provide personalized recommendations that guide consumers toward selected products.¹ These recommendations already have a substantial influence on consumption choices (Ricci et al., 2022), and their importance is likely to increase as technology improves and platforms keep accumulating data.

Emilio Calvano: ecalvano@luiss.it

Giacomo Calzolari: giacomo.calzolari@eui.eu

Vincenzo Denicolò: vincenzo.denicolo@unibo.it

Sergio Pastorello: sergio.pastorello@unibo.it

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¹Although we focus on product recommendations to consumers, RSs are also used to suggest financial products, match workers with firms, curate social-media feeds, recommend academic articles or referees, etc.

Algorithmic recommendations have a natural direct benefit: they improve matching between consumers and products. At the same time, by directing different consumers toward different products based on estimated match values, they reshape the search process and create *personalized prominence*: each consumer begins her search with a product selected specifically for her. This changes the demand faced by firms and thereby affects product-market competition and equilibrium prices.

The main objective of the paper is to analyze this mechanism and its equilibrium effects on firms and consumers. To this end, we develop a framework that embeds realistic RSs in a model of consumer search and price competition. The framework has three key ingredients. First, we implement recommendation algorithms that resemble those used in practice. Second, we generate the sparse and noisy feedback data on which these algorithms are trained, so that they operate with information comparable to that available to real-world platforms. Third, we introduce a model of consumer search in which recommendations determine which product consumers inspect first, while leaving them free to continue searching. We then study equilibrium pricing and welfare in this environment, comparing markets with personalized recommendations to a benchmark of unassisted search.

This approach requires departing from standard search models with purely idiosyncratic preferences. Such models are ill-suited to the study of RSs: if valuations are independent across consumer-product pairs, past ratings contain no information about untried products, and algorithmic recommendations cannot be informative. RSs are valuable precisely because they exploit systematic structure, using similarities and dissimilarities among consumers and among products to predict unobserved valuations. We therefore specify preferences with a systematic component, generated by consumer tastes and product characteristics, and an idiosyncratic component. This gives the algorithm meaningful patterns to learn while leaving recommendations noisy and imperfect.

The use of realistic RSs precludes a complete analytical solution of the model. We therefore rely on numerical experiments on synthetic markets. This approach allows us to characterize equilibrium outcomes while controlling the quantity and quality of the information available to the algorithm. To address issues of external validity, we conduct an extensive robustness analysis to examine how the results vary with the number of products, search costs, data sparsity, noise, and the recommendation technology.

We find that algorithmic recommendations lead to systematically higher prices. Three distinct mechanisms drive this result. The first is a *front-loading effect*. Since search is costly, consumers who inspect a product first are partially captive to the firm supplying that product. Under random search, these consumers are a random draw from the population. Under personalized prominence, by contrast, they are disproportionately consumers with high match values for that product, while low-match consumers are directed towards other products, which are better matches for them. Recommendations, therefore, increase demand from consumers with high willingness to pay and reduce demand from consumers with low willingness to pay. This lowers demand elasticity in the relevant range and induces firms to raise prices.

The second mechanism is the *reliance effect*. When consumers recognize that recommendations are informative, they infer that non-recommended products are, on average, less attractive than randomly drawn products. This lowers the value of continued search and makes consumers more captive to the first firms they visit. The magnitude of this effect depends on consumers' assessment of the informativeness of recommendations. We consider two polar cases: *conservative beliefs*, under which consumers do not update their prior distribution of match values and the reliance effect is absent, and *rational expectations*, under which they correctly infer the distribution of match values of non-recommended products.

Under rational expectations, the reliance effect can raise prices substantially more than the front-loading effect alone. As a result, algorithmic recommendations may reduce consumer

surplus despite improving matching and lowering search costs. Under conservative beliefs, by contrast, the resulting price increase generally does not outweigh these benefits. Somewhat paradoxically, consumers may therefore be harmed precisely because they trust the recommendations.

Even under conservative beliefs, however, consumer surplus need not rise monotonically as algorithms obtain more and better data. Better information helps consumers find products that better match their tastes, but it also strengthens the sorting of high-match consumers and thereby raises prices. The matching benefit is subject to diminishing returns, whereas the price effect becomes stronger as recommendations become more precise. As a result, consumer welfare may decline once recommendations become sufficiently precise.

The front-loading and reliance effects operate in Wolinsky–Anderson–Renault type search models (Wolinsky, 1986, Anderson and Renault, 1999), in which consumers observe neither individual prices nor match values before visiting a firm and therefore search randomly. The third mechanism, which we call the *price-salience effect*, arises when prices are observed before inspection. In that case, search is directed even without recommendations: the order in which consumers inspect products depends on prices. Firms therefore have an additional incentive to cut prices, since a lower price can both increase the probability of purchase conditional on inspection and move the product earlier in consumers' search order. This further undercutting motive exists with or without RSs, but it is weaker with RSs. The reason is that recommendations rank products using estimated match values as well as prices, so price is only one component of prominence. A price cut therefore has less influence on search order than it would under unassisted search. RSs thus soften this channel of price competition and put further upward pressure on equilibrium prices.

These results qualify the common view that recommendations can harm consumers only when they are manipulated to favor particular products, as in self-preferencing. Our final contribution studies precisely this practice and shows that strategic manipulation can actually lower prices relative to genuine recommendations. The effect is especially strong under rational expectations, because manipulation reduces the informational content of recommendations and thereby attenuates the reliance effect. The resulting price response also limits the profitability of manipulation, so the profit-maximizing degree of manipulation may be relatively modest. In our setting, self-preferencing is therefore better understood as a potentially exclusionary practice, because it harms rival firms, than as an exploitative one, since consumers may benefit.

The literature on the economic effects of algorithmic recommendations for final consumers is still small.² Kaye (2025) considers personalized recommendations as a counterfactual in his empirical study of the online hotel market. Consistent with our findings, he concludes that personalized recommendations would raise prices and could reduce consumer surplus. Unlike our analysis, however, he does not explicitly model the recommendation technology; instead, he uses observed data to predict the recommendations the platform might make. Lee and Wright (2026) and Castellini et al. (2023) are closer to our paper in that they specify the RSs explicitly and study their behavior through numerical simulations. However, they do not embed RSs in a model of individual search and analyze their effects on product-market competition.

The rest of the paper is divided into two parts. The first part presents the analytical framework. Section 2 introduces a model of product differentiation with systematic similarities and differences among consumers and products. Section 3 provides a self-contained description of the recommendation algorithms we use. These algorithms are chosen to be consistent with the structure of preferences, so that they are not misspecified relative to the product-market environment. Section 4 describes the data used to train the algorithms and the training procedure.

²We discuss the related search literature in the sections where the relevant mechanisms are introduced.

Section 5 embeds recommendations in a model of consumer search. Section 6 specifies the baseline parameterization used in the numerical analysis.

The second part presents our results. Section 7 examines the value of recommendations when prices are fixed, focusing on better matching and lower search costs. Section 8 starts the analysis of the effect of RSs on equilibrium prices by focusing on the front-loading effect. Section 9 varies the quantity and quality of information available to the algorithm and shows that the relationship between algorithmic information and consumer surplus can be inverted-U shaped. Section 10 analyzes the reliance effect, which arises when consumers update their continuation value from search after receiving an informative recommendation. Section 11 studies directed search and introduces the price-salience effect. Section 12 turns to manipulated recommendations and self-preferencing. The final section discusses extensions and policy implications. Additional details are provided in the online appendix.

2. PRODUCTS AND PREFERENCES

We consider a single platform that intermediates between I buyers, indexed by $i = 1, \dots, I$, and the sellers of J products, indexed by $j = 1, \dots, J$.

Consumers' preferences over all of these products exhibit systematic similarities and differences. This structure is essential for the analysis of RSs: without it, past observations would contain no meaningful patterns that an RS could exploit to predict match values for consumer-product pairs that have not yet been observed. However, the J products need not all be direct substitutes. Instead, we partition them into distinct product markets and assume that each consumer purchases at most one product in each market. Products within a market are therefore substitutes, whereas products in different markets enter separate purchase decisions and do not compete directly for the same demand.

It is nevertheless necessary to model the full set of products jointly because consumers' valuations are correlated across markets. Behavior in one market may therefore reveal information about a consumer's preferences in another. For example, a platform may offer both books and movies: consumers who enjoy war films may also tend to prefer historical novels, while those who enjoy wildlife documentaries may be more likely to value science books. Such cross-market correlations improve the algorithm's ability to infer consumer preferences.

Specifically, we assume that consumer i 's willingness to pay for product j , measured in monetary units, is given by

$$u_{ij} = \bar{u}_{ij} + \varepsilon_{ij}, \quad (1)$$

plus a constant large enough that markets are always covered. The first term, $\bar{u}_{ij}$, is the systematic component of the match value. We specify it as:

$$\bar{u}_{ij} = \sum_{h=1}^H t_{ih} v_{jh} = \mathbf{t}_i \cdot \mathbf{v}_j. \quad (2)$$

The parameter H can be interpreted as the number of product dimensions that matter for consumers. The vector $\mathbf{v}_j = (v_{j1}, \dots, v_{jH})$ describes the position of product j along these latent dimensions, while $\mathbf{t}_i = (t_{i1}, \dots, t_{iH})$ describes consumer i 's tastes over them. These dimensions can be thought of as product characteristics.³ We do not restrict the coordinates t_{ih} and v_{jh} to be non-negative.

³Thus, this specification can be viewed as a Lancasterian characteristics model, in which consumers derive "satisfaction from characteristics that [...] cannot be purchased directly, but are incorporated in goods" (Lancaster et al., 1974, p. 567). It can also be interpreted as a spatial model of product differentiation, as shown below.

The second term, ε_{ij} , is an idiosyncratic shock. We assume that the shocks are i.i.d. across consumers and products and follow a Gumbel distribution with parameters $(-\gamma\beta, \beta)$ where γ is the Euler-Mascheroni constant, so that the mean is zero. These shocks serve to smooth demand and guarantee the existence of a pure-strategy price equilibrium.

We normalize the intensity of consumers' tastes by assuming $\sum_{h=1}^H t_{ih}^2 = 1$ for all consumers i . Consumers therefore differ in their tastes, but not in the overall willingness to pay (i.e., in income). Analogously, we assume that $\sum_{h=1}^H v_{jh}^2 = 1$ for all products j . This assumption makes product differentiation purely horizontal: products differ in the mix of characteristics they embody, but not in their overall quality. This is consistent with our focus on the role of personalized prominence.⁴

Under these normalizations, consumers and products can be represented as points on the unit hyper-sphere $\{\mathbf{x} \in \mathbb{R}^H : \|\mathbf{x}\| = 1\}$. For a consumer of type $\mathbf{t}$, the ideal product is located at $\mathbf{v} = \mathbf{t}$ and yields the maximal systematic match value, $\bar{u} = 1$. Consuming any other product entails a loss of

$$1 - \bar{u}_{ij} = 1 - \mathbf{t}_i \cdot \mathbf{v}_j = \frac{1}{2} \|\mathbf{t}_i - \mathbf{v}_j\|^2.$$

The loss is proportional to the squared Euclidean distance between the consumer and the product on the sphere and is therefore analogous to a transportation cost in a spatial model. The worst match for consumer $\mathbf{t}$ is the antipodal product, $\mathbf{v} = -\mathbf{t}$, which yields $\bar{u} = -1$. The normalization therefore sets the gap between the best and worst systematic match values equal to 2. This implicitly fixes the unit in which monetary values are measured.

It appears that our specification of demand is a standard spatial model of horizontal product differentiation, written in an inner-product form. This type of structured preference environment is rarely used in the search literature, where match values are typically assumed to be entirely idiosyncratic.⁵ For the study of RSs, however, such structure is essential: it is precisely the systematic component of preferences that allows the algorithm to infer match values for consumer-product pairs that have not yet been observed.

To close the model, we assume that consumer and product locations are drawn independently and uniformly from the unit hypersphere. Here we depart from canonical spatial models with equally spaced products. This serves to keep the search problem stationary: in our setting, learning the location of one product conveys no information about the locations of the others, so beliefs about uninspected products remain unchanged during search.

3. RECOMMENDER SYSTEMS

Given the preference structure introduced above, we focus on *factor-based* recommender systems with bilinear scoring rule. This class of algorithms is natural in our setting because its statistical model mirrors the structure of match values in the economy. In the markets we study, a consumer's systematic willingness to pay for a product is the inner product of a consumer taste vector and a product-characteristic vector. Factor-based algorithms impose the same structure: they explain observed ratings, and predict unobserved ones, by estimating user and item vectors whose inner products generate predicted evaluations.

Therefore, these algorithms are correctly specified relative to our economic environment. This is important for interpretation. If the algorithm's statistical model were misspecified relative to the true structure of preferences, the effects of recommendations could depend on the

⁴An extension to vertical differentiation, where $\sum_{h=1}^H v_{jh}^2$ may vary across products, is analyzed in the discussion-paper version of the paper.

⁵Notable exceptions include Wolinsky (1983) and de Cornière and Taylor (2014).

particular form of misspecification. Our modeling choice instead isolates the effects that arise when the RS is well suited to the task, so that recommendations are imperfect only because the available ratings are sparse and noisy.

Factor-based methods are also a natural choice from the perspective of the computer science literature. They played a central role in the Netflix Prize, which helped establish them as a leading approach to personalized recommendations.⁶ Owing to their computational efficiency, scalability, and interpretability, they remain widely used in practice.⁷ They also frequently serve as building blocks in more complex recommendation architectures.⁸

Modern recommender systems can, of course, be more complex, accommodating richer preference structures, contextual variables, multimodal information, etc. (Deldjoo et al., 2024). In real-world applications, this additional flexibility may be valuable.⁹ In our environment, however, the true preferences are exactly bilinear. Any extra flexibility therefore offers no structural advantage: a more complex algorithm trained on the same synthetic data would either be misspecified or approximate the same factor-based relationship in a less transparent manner.

In the rest of this section, we provide a brief, self-contained description of factor-based algorithms. We consider two versions: *latent-factor collaborative filtering*, in which both user and item vectors are estimated from ratings, and *content-based* systems, in which product characteristics are observed by the platform.

3.1. The basic problem

Consider a finite set of users $i = 1, 2, \dots, I$, a finite set of items $j = 1, 2, \dots, J$ and a sparse matrix $\tilde{\mathbf{R}}$ of size $I \times J$, which is commonly referred to as the *rating* matrix in the computer science literature (we adopt this terminology throughout). Existing entries $\tilde{r}_{ij}$ represent the rating for item j either reported by user i or inferred from user behavior, and the entries for the unobserved user-item pairs are missing. The objective of the algorithm is to estimate the full matrix of true ratings, denoted as $\mathbf{R}$. Economists naturally interpret the true ratings r_{ij} as the expected utility user i derives from item j , and the reported ratings $\tilde{r}_{ij}$ as noisy signals of that utility.

The algorithm re-estimates the observed values and fill in the missing ones exploiting the correlation structure of observed ratings. The underlying assumption is that users whose observed ratings are similar are likely to have similar ratings for unobserved items, while items whose ratings are similar across observed users are likely to be rated similarly by unobserved users.

3.2. Factor-based models

The factor-based algorithms we focus on explain the observed user-item ratings and predict the unobserved ones by using observable or hidden (latent) factors that represent inherent properties of both items and users. This approach reduces the dimensionality of the rating matrix by representing the ratings in terms of a relatively small number of variables.

⁶In 2006, Netflix announced a one-million-dollar prize for the first recommender system that could improve the accuracy of Cinematch, its algorithm for predicting ratings, by at least 10 percent. The prize was awarded in 2009 to BellKor’s Pragmatic Chaos, whose solution improved prediction accuracy by 10.06 percent. A subsequent challenge was discontinued because of concerns that the anonymity of the data could be compromised. See Koren et al. (2009).

⁷See, for example, (Ricci et al., 2022, Koren et al., 2022, Rendle et al., 2020, 2022).

⁸See, for example, (He et al., 2020, Zhang et al., 2019).

⁹Its incremental value relative to simpler factor-based methods may nevertheless be limited. A team of Netflix researchers has found that replacing Netflix’s production recommender with a bilinear factor-based system would reduce engagement by only 4 percent (Zielnicki et al., 2026).

Analytically, the true rating of item j by user i is viewed as the inner product of a vector of user-specific parameters $\mathbf{t}_i$ and a vector of item-specific parameters $\mathbf{v}_j$:

$$r_{ij} = \sum_{h=1}^H t_{ih} v_{jh}, \quad (3)$$

where H is the number of factors that the algorithm posits. In matrix notation, $\mathbf{R} = \mathbf{T}\mathbf{V}'$, where $\mathbf{T}$ and $\mathbf{V}$ are the $(I \times H)$ and $(J \times H)$ matrices formed by the I vectors $\mathbf{t}_i$ and the J vectors $\mathbf{v}_j$, respectively. The H factors may have a semantic interpretation. For the algorithm, however, they need not have any specific meaning.

3.3. Estimation procedure

Within the broad class of algorithms described above, we consider two different specifications. In *latent-factor, collaborative-filtering* RSs, "attributes" and "tastes" are estimated jointly. The algorithm estimates the parameters $\mathbf{T}$ and $\mathbf{V}$ by minimizing some measure of the distance between the estimated ratings,

$$\hat{r}_{ij} := \sum_{h=1}^H \hat{t}_{ih} \hat{v}_{jh} \quad (4)$$

and the observed ones, $\tilde{r}_{ij}$. Using the Euclidean distance, the estimated parameters $\hat{\mathbf{T}}$ and $\hat{\mathbf{V}}$ solve:

$$\min_{\hat{\mathbf{T}}, \hat{\mathbf{V}}} \sum_{(i,j) \in S} \left(\tilde{r}_{i,j} - \sum_{h=1}^H \hat{t}_{ih} \hat{v}_{jh} \right)^2, \quad (5)$$

where S denotes the set of all pairs (i, j) for which r_{ij} is (imperfectly) observed.¹⁰ Using the completed matrix $\hat{\mathbf{R}} = \hat{\mathbf{T}}\hat{\mathbf{V}}'$, the algorithm can generate rankings of all items for all users, allowing it to provide personalized recommendations.

Alternatively, in *content-based* RSs, it is assumed that the algorithm can observe the attributes (i.e., the v_{jh} s) and must estimate only tastes (i.e., the t_{ih} s). This is more similar to the approach taken in most empirical demand estimations based on the Random Utility Model.¹¹ In this case, the estimated parameters $\hat{\mathbf{T}}$ solve:

$$\min_{\hat{\mathbf{T}}} \sum_{(i,j) \in S} \left(\tilde{r}_{i,j} - \sum_{h=1}^H \hat{t}_{ih} v_{jh} \right)^2. \quad (6)$$

3.4. Regularization

In real-life applications, the density d of the matrix of observed ratings $\tilde{\mathbf{R}}$ (i.e., the fraction of its non-empty cells) is typically very low, often below 2% (Chin et al. (2022)). Estimates

¹⁰Problem (5) assumes that the algorithm uses a pre-specified number of latent factors H , as we do in our baseline analysis. However, in a robustness check, we allowed the algorithm to estimate the value of H using an internal cross-validation routine, as described in Section A.3 of the online appendix. Results remain unchanged.

¹¹If only some product attributes are observed by the RS, one can use hybrid systems that combine content-based and collaborative-filtering methods. (By symmetry, the same approach can accommodate partially observed consumer tastes.) Since, as shown below, the two pure systems generate very similar results in our setting, we do not consider hybrids separately.

based on such sparse data may therefore suffer from small-sample problems. (The small-sample properties of factor-based estimators are difficult to characterize analytically, which is one reason why the analysis of these RSs often relies on numerical methods.) In particular, the estimation procedure (5) (or (6)) may cause overfitting. A common approach for addressing this problem is to use regularization techniques. The idea is to penalize large values of the coefficients in the matrices $\hat{\mathbf{T}}$ and $\hat{\mathbf{V}}$ so as to decrease the variance of $\hat{\mathbf{R}}$. The actual minimization problem then becomes:

$$\min_{\hat{\mathbf{T}}, \hat{\mathbf{V}}} \sum_{ij \in S} \left(\tilde{r}_{i,j} - \sum_{h=1}^H \hat{t}_{ih} \hat{v}_{jh} \right)^2 + \lambda_t \sum_{i=1}^I \sum_{h=1}^H \hat{t}_{ih}^2 + \lambda_v \sum_{j=1}^J \sum_{h=1}^H \hat{v}_{jh}^2, \quad (7)$$

where the non-negative parameters λ_t and λ_v are the regularization weights. We choose the weights using a cross-validation procedure described in Section A of the online appendix.

4. TRAINING DATA

After purchasing a product, consumers may explicitly or implicitly report a rating to the platform. These ratings are the data used by the algorithms to estimate the match values.

In the baseline scenario, we assume that the reported rating $\tilde{u}_{ij}$ is equal to the utility actually experienced, u_{ij} , plus a noise term:

$$\tilde{u}_{ij} = u_{ij} + \epsilon_{ij}. \quad (8)$$

The reporting noise ϵ_{ij} is normally i.i.d. with zero mean and variance σ_ϵ^2 . The higher the variance σ_ϵ^2 , the less informative the ratings reported by consumers.

Later on, we will consider extensions where the report is made on a Likert scale. It is important to note that ratings on a two-level Likert scale may be available to a platform even without explicit reporting by users. For instance, the platform could monitor whether a consumer has purchased an inspected product or watched a movie until the end.

As noted, RSs typically have access to information about only a small fraction of all consumer-product pairs. What determines which ratings are observed and which are not? We consider two alternative data-generating processes, depending on whether the data have been generated based on recommendations provided by the algorithm itself in the past or not. In our baseline analysis, we assume that the algorithm itself does not play a role in generating the data it uses. Specifically, we assume that for each user, the platform observes ratings for a fixed number of products, drawn randomly and independently across users.

Alternatively, we consider the case where the ratings observed by the platform have been created based on recommendations provided by the algorithm itself in the past. Details about this process are provided in Section B of the online appendix. Data endogeneity creates a feedback loop that has been extensively discussed in the computer science literature: the algorithm gathers better information on the user-item pairs it has selected in the past.¹² This implies that even if the final density matches that of random data, the user-item pairs observed by the algorithm are somewhat correlated, diminishing the quality of the available information. Consequently, the precision of estimates and the quality of recommendations decline.

¹²This is often referred to as the “bias in the algorithms” (even though a more precise designation might be “bias in the data”).

5. SEARCH

We now embed the preference structure and recommendation technology described above in a model of consumer search. We interpret recommendations as a form of *personalized prominence*: they provide credible pre-search information that affects the order in which consumers inspect products. Consumers are not assumed, however, to purchase the recommended product mechanically. If it proves unattractive upon inspection, they remain free to continue searching.

While the RS is trained using data on the entire set of products, search and price competition are market-specific. We therefore henceforth focus on a single market with m products and denote the set of these products as M . As explained above, products in a market M compete for the same purchase; products in different markets do not but still provide information that the RS uses to estimate the match values.

5.1. Prior information

Consumer search depends on what consumers know before inspecting products. In general, consumers may have prior information about product availability, prices, and match values. Since the specific role of RSs is to provide information about match values, we isolate their effect by holding other forms of information fixed.

Our baseline follows the search environment of Wolinsky (1986) and Anderson and Renault (1999): consumers know which products are available and understand the distribution of product characteristics and equilibrium prices, but they do not observe individual prices or match values before inspection. Inspecting a product is costly. Upon inspection, the consumer observes both the product's price and her match value with that product.

In this setting, we assume that recommendations are based only on the algorithm's estimated match values, not on prices. This cleanly separates match-value information from price information: if recommendations were allowed to depend on prices while consumers did not otherwise observe prices, the recommendation would also transmit price information, confounding the channel that we want to study.

In Section 11, we consider an extension in which consumers observe prices before inspection. Search is then directed, and consistency requires recommendations to rank products by estimated net surplus rather than by match values alone. This introduces the price-salience effect mentioned in the introduction, which is absent from the baseline environment.

5.2. Random search benchmark

When prices are not observable, in the benchmark without recommendations, consumers have no product-specific information before search. Let $s_{ij} = u_{ij} - p_j$ denote the surplus that consumer i obtains from product j , where p_j is the price. Before inspection, the surplus from any product is distributed according to a common prior F_s . This prior is induced by three objects: the distribution $F_{\bar{u}}$ of the systematic match-value component $\bar{u}_{ij}$, the distribution F_{ε} of the idiosyncratic shock ε_{ij} , and the equilibrium distribution of prices F_p .

By symmetry of consumers and products, this prior is the same for all consumers and all products. Since no product is distinguished ex ante, consumers sample products at random. Each inspection entails a search cost c_s . Upon inspecting a product, the consumer observes its price and match value, and hence its realized surplus.

Standard search arguments imply a reservation-surplus rule. Let $\hat{s}$ denote the cutoff level of the surplus. Using the notation $[x]_+ = \max\{x, 0\}$, in the unassisted-search benchmark the cutoff is implicitly defined by

$$\int \int \int [\bar{u} + \varepsilon - p - \hat{s}]_+ dF_{\bar{u}}(\bar{u}) dF_{\varepsilon}(\varepsilon) dF_p(p) = c_s. \quad (9)$$

The left-hand side is the expected gain from one additional inspection when the consumer's current best surplus is $\hat{s}$. Condition (9) states that at the optimum, this must be equal to the search cost c_s . Consumers therefore stop searching once they have observed a surplus at least as large as $\hat{s}$, and continue otherwise.

The rule is stationary because the surplus distribution generated by the next inspection is independent of the products already inspected. This follows from the independent-location assumption: learning one product's systematic match value, and hence its location, conveys no information about the locations or match values of the remaining products.

5.3. Personalized prominence

We now introduce personalized recommendations.¹³ Let $\hat{u}_{ij}$ denote the algorithm's estimate of the match value between consumer i and product j . In the baseline model, where prices are not observed before inspection, the recommendation is based, for consistency, on these estimated match values alone. For now, we assume that the platform recommends a single product:

$$r_i \in \arg \max_{j \in M} \hat{u}_{ij}. \quad (10)$$

Later, we consider extensions in which the platform recommends a list of products or manipulates recommendations, so that the recommended product need not be that defined in (10).

The recommendation changes the consumer's first inspection. Instead of sampling a product at random, consumer i first inspects the product recommended to her. Given the lack of other information, it is rational for the consumer to do so as soon as she believes that the recommendation is at least minimally informative. Upon inspection, the consumer observes the price p_{r_i} and learns her match value u_{ir_i} , and hence the surplus $s_{ir_i} = u_{ir_i} - p_{r_i}$. She then decides whether to buy the recommended product or to continue searching. Any subsequent search is random, as in the benchmark.

There are two ways to model the consumer's continuation decision after the recommended product has been inspected. The first assumes that the consumer uses the recommendation only to decide where to start searching, but does not otherwise revise her continuation value from search (except that she anticipates the new equilibrium price distribution). This case captures the front-loading effect of recommendations described in the introduction: the first product visited by a consumer is more likely to be high value.

The second case allows for rational reliance on the recommendation. If consumers understand that recommendations are informative, they also understand that products not recommended to them are, on average, less attractive than randomly drawn products. The distribution of surplus from continued search therefore differs from the ex ante distribution: after receiving a recommendation, consumers condition on the fact that the remaining products were not selected by the RS. This posterior distribution is shifted downward relative to the prior. The expected return from additional search falls, consumers use a lower cutoff, and thus they are more willing to buy the recommended product. This generates the reliance effect. Initially, we focus on the front-loading effect; the reliance effect is analyzed in Section 10.

¹³In our environment, where product differentiation is purely horizontal and no product is systematically preferred by all consumers, non-personalized recommendations are of limited value; see footnote 23 and Section D of the online appendix. They could play a more meaningful role in a setting with vertical differentiation.

5.4. Product-market competition

Each product is supplied by a different single-product firm. Firms compete in prices. Marginal production costs are normalized to zero, so prices can be interpreted as price-cost margins.

We model price competition as a complete-information game, assuming that each firm observes the locations of its competitors' products. This implies that even though firms are symmetric *ex ante*, they become heterogeneous once product locations are realized and may therefore charge different equilibrium prices.

One might assume that firms do not observe individual consumer locations and therefore compute expected demand by integrating over both the systematic match values, $\bar{u}_{ij}$, and the idiosyncratic shocks, ε_{ij} . Because the number of consumers is large, however, this is effectively equivalent to conditioning on the realized distribution of systematic match values and integrating only over the idiosyncratic shocks to ensure smoothness. For computational convenience, we adopt this latter formulation.

5.4.1. Demand with random search

We now derive the demand functions faced by the firms (see Section C of the online appendix for further details).

Since the market is covered, each consumer eventually purchases one of the m products in the market. In the unassisted-search benchmark, the probability that consumer i purchases product j is:

$$X_{ij} = q_{ij} \left[\sum_{z=0}^{m-1} \frac{1}{1+z} B(z; \mathbf{q}_{i,-j}) \right] + \prod_{s \in M} (1 - q_{is}) \times \frac{e^{\frac{\bar{u}_{ij} - p_j}{\beta}}}{\sum_{s \in M} e^{\frac{\bar{u}_{is} - p_s}{\beta}}}. \quad (11)$$

The first term is *fresh demand*: demand from consumers who purchase product j immediately upon inspection because it is the first *acceptable* product they encounter. Product j is acceptable to consumer i if its realized surplus exceeds the search cutoff: $\bar{u}_{ij} + \varepsilon_{ij} - p_j \geq \hat{s}$ or, equivalently, $\varepsilon_{ij} \geq \hat{s} + p_j - \bar{u}_{ij}$. From the viewpoint of firm j , which observes $\bar{u}_{ij}$ but not the realization of ε_{ij} , the probability of this event is therefore $q_{ij} = 1 - F_\varepsilon(\hat{s} + p_j - \bar{u}_{ij})$. Under random search, if product j is acceptable and exactly z other products are also acceptable, then each of the $1 + z$ acceptable products is equally likely to be the first acceptable product inspected. Product j is therefore purchased with probability $1/(1 + z)$ conditional on this event. Let $\mathbf{q}_{i,-j} = (q_{i1}, \dots, q_{i,j-1}, q_{i,j+1}, \dots, q_{im_M})$ denote the vector of acceptability probabilities for all products other than j , and let $B(z; \mathbf{q}_{i,-j})$ be the probability that exactly z of these products are acceptable. That is, $B(\cdot; \mathbf{q}_{i,-j})$ is the Poisson-binomial distribution generated by $\mathbf{q}_{i,-j}$. Summing over all possible values of z gives the fresh-demand term in (11).

The second term in (11) is *returning demand*. This is demand from consumers who inspect all products without finding any product whose realized surplus exceeds the cutoff, and then return to the product that delivers the highest realized surplus. The probability that no product is acceptable to consumer i is $\prod_{s \in M} (1 - q_{is})$. Conditional on this event, the consumer buys the product with the highest realized surplus. Because the idiosyncratic shocks are Gumbel-distributed, the probability that product j delivers the highest surplus is given by the standard logit formula.

Expected demand for product j is obtained by summing purchase probabilities across consumers: $D_j(\mathbf{p}) = \sum_{i=1}^I X_{ij}$. Prices enter demand in two ways: through the acceptability probabilities q_{ij} , which determine fresh demand, and through the surplus comparison that determines returning demand.

5.4.2. Demand with recommendations

Now suppose that each consumer i receives a recommendation for a product, $r_i \in M$, and inspects that product first. We assume that firms observe the recommendation assignment.¹⁴

Recommendations affect demand by changing the order in which products are inspected. The fresh demand term becomes:

$$\mathbf{1}\{j = r_i\}q_{ij} + \mathbf{1}\{j \neq r_i\}(1 - q_{ir_i})q_{ij} \sum_{z=0}^{m-2} \frac{1}{1+z} B(z; \mathbf{q}_{i,-\{j,r_i\}}) \quad (12)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. If $j = r_i$, product j is inspected first and is purchased immediately whenever it is acceptable, so its fresh-demand probability is simply q_{ij} . If instead $j \neq r_i$, product j can generate fresh demand only if the recommended product is not acceptable, an event that occurs with probability $1 - q_{ir_i}$. Conditional on this event, search continues randomly among the remaining products. Product j is then purchased as fresh demand if it is acceptable and if it is the first acceptable product inspected among the $m - 1$ products other than r_i . This gives the second term in (12).

The expression for returning demand is the same as in (11). The reason is that returning demand arises only when the consumer inspects all products, in which case the order of inspection is irrelevant. As before, expected demand for product j is obtained by summing purchase probabilities across consumers.

Besides changing the expression for fresh demand, recommendations can also affect demand indirectly through the search cutoff $\hat{s}$, and therefore through the acceptability probabilities q_{ij} . The cutoff may change because recommendations alter equilibrium prices, and hence the distribution of surplus from continued search. It may also change if consumers infer from the recommendation that non-recommended products have a different distribution of match values from randomly drawn products.

5.5. Pricing equilibrium

Given the demand functions, and with zero marginal costs, the Bertrand-Nash equilibrium prices $\mathbf{p}^*$ satisfy

$$p_j^* \in \arg \max_{p_j} p_j D_j(p_j, \mathbf{p}_{-j}^*) \quad \text{for all } j \in M.$$

We compute the equilibrium numerically by an iterative procedure that solves the system of first-order conditions associated with firms' profit maximization.¹⁵ Starting from an initial vector of prices $\mathbf{p}_0$, at each iteration $\tau = 1, 2, \dots$ we compute the search cutoff implied by the current candidate prices,¹⁶ construct the corresponding demand functions, and then update each

¹⁴If firms had only coarser information, such as the probability with which each product is recommended to each consumer, demand would be obtained by averaging the expressions below over this additional uncertainty.

¹⁵To solve the system of first-order conditions, we use subroutine DNQSOL in the Caltech MATH77 Fortran library, exercising special care in checking the second-order conditions.

¹⁶Since the calculation of the cutoff is time consuming, to speed up the calculation we compute the cutoff using the true distribution of product characteristics and idiosyncratic shocks but a degenerate prior over prices, as if consumers expected all firms to charge the same price, equal to the true average price. This simplification abstracts from the option value generated by price dispersion and therefore understates the value of continued search, as would an increase in the search cost. Since the same cutoff procedure is used in both the unassisted-search benchmark and the RS case, this approximation has little practical effect on the comparison between the two regimes. We have verified this by using the true distribution of equilibrium prices in a few pilot sessions.

firm's price. For each product j , the new price $p_{j,\tau}$ maximizes firm j 's profit holding the other prices and the cutoff fixed.¹⁷ This yields a new candidate price vector $\mathbf{p}_\tau$. The procedure is iterated until convergence.

6. BASELINE PARAMETRIZATION

In this section, we specify the baseline parameterization of the model for the numerical analysis. This specification serves as the base for the extensive robustness analysis reported in Section 9 and the online appendix.

Whenever possible, we choose baseline values that match the Netflix Challenge (see footnote 6 and Table I).¹⁸ In the Netflix dataset, the ratio of users to items was approximately 30, so we set $I/J = 30$. To reduce the computational burden, however, we scale down both dimensions by a factor of approximately 20 relative to the Netflix problem. We therefore set $J = 800$ and $I = 24,000$. The density d of the observed rating matrix $\tilde{\mathbf{R}}$ was 1.2%, and we adopt the same value in the baseline.

	Environment				Recommender System			
	Users I	Items J	Observations	Density	H	ℓ	H	ℓ
Netflix	500,000	17,000	100,000,000	1.2%	40	4.84	100	1.93
Baseline	24,000	800	230,400	1.2%	3	3.10	5	1.86

TABLE I: The baseline scenario and the Netflix Challenge

The winner of the Netflix Challenge used a number of latent factors H of either 40 or 100. (The trade-off is that more latent factors allow for more flexibility but increase the number of variables to be estimated with the same observations, reducing the precision of the estimates.) To identify comparable values for our parametrization, we match the ratio of observations to estimated parameters,

$$\ell = \frac{d}{H} \frac{J \times I}{(J + I)}. \quad (13)$$

This criterion points to two values of H : $H = 2$, which implies $\ell = 4.65$ and corresponds approximately to $H = 40$ in the Netflix Challenge, and $H = 5$, which we consider in the robustness analysis and corresponds to $H = 100$. We do not use $H = 2$ as our baseline, however, because under the uniform-location assumption, it yields a distribution of the systematic match value, $\bar{u}_{ij} = \mathbf{t}_i \cdot \mathbf{v}_j$, with substantial mass near the endpoints -1 and 1 . We instead set $H = 3$, for which $\bar{u}_{ij}$ is uniformly distributed on $[-1, 1]$.

For parameters on which the Netflix Challenge and similar datasets provide no guidance, we proceed as follows. First, we report results for markets with $m = 7, 13$ and 19 products. We have also considered a broader and more finely spaced grid of values, but this yielded no additional insights, as the results vary monotonically with m .

¹⁷This is consistent with the assumption that consumer search decisions depend on expected prices rather than on deviations from expected prices, an assumption often referred to in the literature as *passive beliefs*.

¹⁸Computer scientists also routinely use other benchmark datasets to evaluate RSs, such as the MovieLens 10M Dataset, which contains movie ratings collected by the GroupLens research project at the University of Minnesota. The corresponding parameter values are not substantially different from those of the Netflix Challenge and lie within the ranges considered in our robustness analysis.

Second, we set the unit search cost c_s at 0.04. To give this parameter a practical interpretation, it implies that the conversion rate from recommendations to purchases (i.e., the share of consumers who purchase the recommended product without searching further) ranges from 55% with $m = 7$ to 78% with $m = 19$. These values appear plausible for relatively inexpensive products, say those priced in the range of a few dozen dollars.¹⁹ For more valuable products, consumers are likely to search more extensively.²⁰ In our model, this corresponds to a lower search cost relative to product values.²¹

Third, we choose the scale parameter β , which governs the dispersion of the idiosyncratic taste shocks ε_{ij} , so that demand is just sufficiently smooth for the pricing game to admit a pure-strategy equilibrium. Specifically, we set $\beta = 0.18$.²² This value implies that the standard deviation of the idiosyncratic taste shocks equals 40% of the standard deviation of the systematic component of match values. The relative dispersion of these two components will play an important role in the analysis of directed search below.

Finally, we set the standard deviation of the reporting noise, σ_ϵ , equal to 20% of the standard deviation of systematic match values. As with the other parameters, we allow this ratio to vary substantially in the robustness analysis.

For each parameter configuration, we run 100 simulations, drawing new realizations of the idiosyncratic taste shocks ε_{ij} , the reporting noise ϵ_{ij} , the random search order, and the set of observed entries in the rating matrix $\tilde{\mathbf{R}}$. In the tables below, we report averages across simulations, with the corresponding standard errors in parentheses.

7. THE VALUE OF RECOMMENDATIONS

Before turning to the main focus of the paper, namely the impact of RSs on equilibrium prices, we first ask how informative the recommendations are and how much value they create for consumers. To this end, we temporarily set all prices equal to marginal cost (i.e., zero) and study the effect of recommendations on consumer surplus.

m	Absolute gains			Percentage gains		
	7	13	19	7	13	19
Collaborative filtering RS	0.0810 (0.0008)	0.1657 (0.0013)	0.2236 (0.0011)	32.18 (0.40)	48.65 (0.49)	58.06 (0.36)
Content-based RS	0.0822 (0.0008)	0.1688 (0.0013)	0.2285 (0.0011)	32.65 (0.40)	49.57 (0.49)	59.32 (0.36)

TABLE II: Changes in consumer surplus at constant prices.

¹⁹Direct evidence is scarce, but several studies point to limited search for relatively inexpensive products. For example, on Amazon, 95% of consumers consider only the featured (Buy Box) offer (Lee and Musolff, 2026); about 75% of online book buyers visit a single store (De los Santos et al., 2012); and consumers click on average only 1.14 hotels per search session on Expedia (Ursu, 2018).

²⁰See, for example, (Kim et al., 2010) for evidence from consumer electronics.

²¹Section F.1 of the online appendix shows that varying search costs does not alter the price effects of RSs, but does affect their magnitude: the lower the search cost, the smaller the effects. In the limit as c_s approaches zero, consumers inspect all products before purchasing, and recommendations no longer affect market outcomes.

²²With this value, for the baseline parametrization all sessions converged to a pure-strategy equilibrium under conservative beliefs, and all but one under rational expectations.

Holding prices constant, recommendations benefit consumers in two ways. First, they direct consumers toward products that are more likely to match their preferences. Second, by making good matches arrive earlier in the search process, they reduce the need for costly search.

Table II reports the increase in consumer surplus relative to the random-search benchmark, both in absolute terms and as a percentage of the gain generated by perfectly informative recommendations, i.e., recommendations that always direct each consumer to the product with the highest systematic match value. Consider first the relative gains, which are independent of the conversion into dollars. Despite being trained on sparse and noisy data, the RS captures a substantial share of the potential benefit of perfectly informative recommendations, ranging from about 30% when the market contains 7 products to nearly 60% when it contains 19.

The content-based algorithm performs slightly better than the collaborative-filtering system. This is natural, since the former observes product characteristics, whereas the latter must infer them from the data. The difference is modest, however, suggesting that collaborative filtering recovers much of the relevant product information even from sparse observations. This is because there are far fewer products than consumers, so the algorithm estimates product characteristics quite precisely. The imperfection in recommendations reflects primarily the difficulty of estimating individual preferences.

The absolute gains range from 8 to 22 cents per consumer in model units. To interpret these magnitudes, recall that we normalize the difference between the match values of the best and worst possible products to 2. In a hypothetical market in which the average match value of a consumer's ideal product is \$20, while that of the worst possible match is \$10, one model unit corresponds to \$5. The estimated gains would therefore range from approximately 40 cents to \$1.10 per consumer. With this conversion rate, the search cost would be 20 cents, corresponding to 48 seconds of inspection time when valued at an hourly wage of \$15. The order of magnitude of these values appears plausible for such a market and provide some reassurance that the baseline parametrization generates economically reasonable outcomes.

Section E of the online appendix decomposes the increase in consumer surplus into gains from improved matching and savings in search costs. The latter account for approximately 60 percent of the total gain. This helps explain why the welfare gains are larger in markets with more products: the higher the number of available products, the more extensively consumers must search to identify a good match in the absence of an RS.²³

8. THE FRONT-LOADING EFFECT

We now turn to the main focus of the paper, namely, the impact of algorithmic recommendations on equilibrium prices.

In this section, we isolate a first mechanism by assuming that consumers interpret recommendations conservatively. That is, they use the recommendation to decide where to start their search, implicitly recognizing that the recommended product is more likely than a randomly drawn product to be a good match. However, they do not revise their beliefs about the distribution of match values among products that were not recommended. In case they continue to search after inspecting the recommended product, they therefore use the same distribution of residual match values as in the random-search benchmark. The search cutoff is adjusted only to reflect the new equilibrium price distribution prevailing under recommendations.

²³The online appendix also reports the welfare gains from non-personalized recommendations (Section D). We consider both a factor-based RS constrained to recommend the same product to all consumers and a popularity-based algorithm that recommends the product with the largest market share under random search. In both cases, the welfare gains are very small, confirming that non-personalized recommendations are not particularly informative in an environment with purely horizontal differentiation such as ours.

This assumption of *conservative beliefs* shuts down the reliance effect that arises if consumers understand that an informative recommendation also conveys negative information about non-recommended products. We analyze this effect in Section 10.

8.1. Prices under conservative beliefs

Even under conservative beliefs, recommendations are in fact informative. The recommended product has, on average, a higher match value for the consumer than a randomly drawn product. RSs, therefore, change the composition of initial visitors. Each firm is more likely to be visited by consumers with high match values for its product, whereas consumers with low match values are more likely to be directed elsewhere. We call this sorting of consumers toward products they value highly, the *front-loading effect*.

The sorting of consumers raises equilibrium price-cost margins. The reason is simple: a larger share of high-value consumers among initial visitors shifts demand upward at high prices, while a smaller share of low-value consumers shifts demand downward at low prices. This makes demand less elastic over the relevant range, inducing firms to raise prices.

Table III reports the percentage changes in equilibrium price-cost margins relative to the unassisted-search benchmark generated by the front-loading effect. Since firms are heterogeneous, the reported values refer to average prices, weighted by market shares.

m	7	13	19
Collaborative filtering RS	12.23 (0.35)	25.38 (0.50)	37.04 (0.64)
Content based RS	13.14 (0.31)	28.70 (0.45)	44.95 (0.55)

TABLE III: Percentage changes in equilibrium prices.

The price increase is significant and becomes larger as the number of products grows. Intuitively, a richer choice set gives the algorithm more scope to sort consumers toward products they value highly. This strengthens the front-loading effect. The price increase is also larger under the content-based algorithm. Content-based recommendations are more informative and therefore generate a stronger front-loading effect.²⁴

8.2. A stylized example

To better understand how the front-loading effect changes the demand function, consider the following simple example, illustrated in the left-hand panel of Figure 1. There are three products, $j = 1, 2, 3$, and three corresponding equally sized groups of consumers. Consumers of type j value product j at 8 plus an idiosyncratic shock uniformly distributed on $[-2, 2]$, so their willingness to pay for their preferred product is uniformly distributed on $[6, 10]$. They value each of the other two products at 4 plus the same type of shock, so their willingness to pay for non-preferred products is uniformly distributed on $[2, 6]$. In the example, the market need not be covered, and captivity created by the first inspection is modeled in a stark way,

²⁴As we show below, this logic is more general: increases in the informativeness of recommendations tend to amplify the front-loading effect, to a large extent regardless of the source of the additional information.

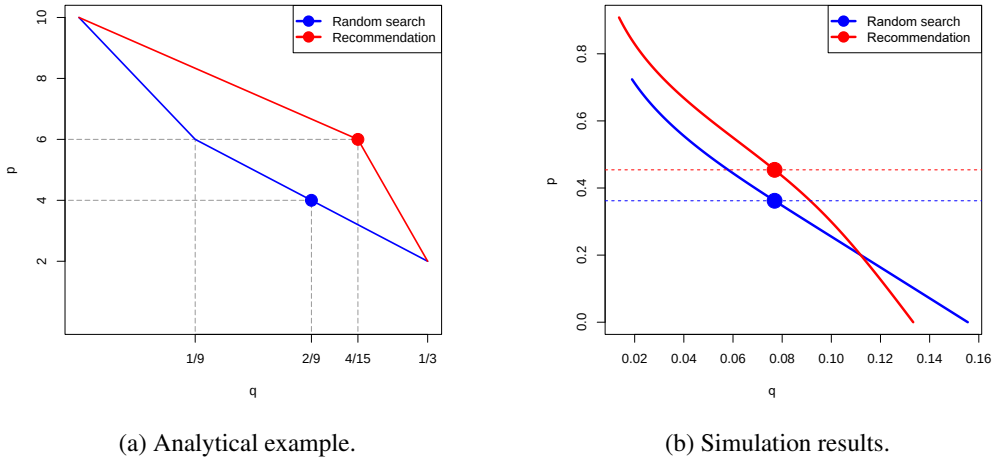


FIGURE 1.—Demand curves and equilibrium prices under random search (blue) and with personalized recommendations (red) in the stylized example (*Left panel*) and in the full model (*Right panel*).

assuming that the first inspection is costless whereas any further search is prohibitively costly. Firms are then local monopolists over their initial visitors.

Under random search, each firm is visited by a representative pool of consumers: one third high-value, regarding the product as their preferred product, and two thirds low-value, regarding it as a non-preferred product. The corresponding demand curve is the blue piecewise-linear curve in Figure 1. It is easy to see that the profit-maximizing price is 4.

Now suppose that an RS directs 80 percent of type- j consumers to product j , while the remaining 20 percent are directed to one of the other products. Each firm is then visited by a much more favorable pool of consumers. The new demand becomes the red curve in Figure 1: its upper part shifts upward and becomes flatter, because the firm now attracts more high-value consumers, while its lower part becomes steeper, because fewer low-value consumers visit the firm. As a result, the profit-maximizing price rises from 4 to 6. At $p = 6$, only high-value visitors are served. Under random search, this price was not optimal because the firm would have an incentive to cut the price in order to sell also to the many low-value consumers who happened to visit it. With the RS, by contrast, the incentive to reduce price in order to serve low-value visitors is much weaker, and the firm optimally focuses on high-value consumers. This shows that algorithmic recommendations effectively segment consumers across firms and increase each firm's local market power even in the absence of price discrimination.

8.3. Demand shifts in the full model

The same mechanism operates in the full model, although the demand shift is more complex because search costs are finite, firms compete for consumers who continue searching, and the market is covered (see [Johnson and Myatt \(2006\)](#) for a comprehensive analysis of demand shifts). The right-hand panel of Figure 1 plots the demand curve for a randomly drawn firm in one of our sessions. As in the stylized example, the upper part of the demand curve shifts upward and becomes flatter: conditional on being visited early, the firm is more likely to face consumers with high match values. The lower part of the demand curve shifts downward but differently from the stylized example. With finite search costs, low-match consumers who are

directed to other products may purchase those products before ever reaching the firm in question. As a result, the RS demand lies below random-search demand at low prices. This feature is absent from the local-monopoly example, where each firm sells only to its initial visitors. Around the equilibrium price, however, the relevant effect is the same as in the example: demand becomes less elastic, inducing firms to raise prices.

8.4. Discussion

Although the front-loading mechanism is both intuitive and powerful, it has received relatively little attention in the search literature. The closest antecedent is [Anderson and Renault \(2000\)](#), who study a model in which a fraction of consumers know their match values before searching. In their two-product model, increasing this fraction raises prices. Intuitively, when consumers are more likely to encounter a good match early in the search process, the competitive pressure created by continued search is weakened. A similar force arises under personalized prominence.

It is important to note the difference between personalized prominence and *uniform* prominence, where the same product is made prominent to all consumers. [Armstrong et al. \(2009\)](#) show that uniform prominence raises the weight of fresh demand relative to returning demand for the prominent product. Since fresh demand is more elastic than returning demand, the prominent firm lowers its price.

Personalized prominence also affects the fresh–returning demand mix, but its main effect is to change fresh demand: initial visitors are disproportionately consumers with high match values for the recommended product, so fresh demand becomes less elastic. In our simulations, this mechanism dominates the composition effect emphasized by [Armstrong et al. \(2009\)](#). Thus, personalized prominence can raise prices even though uniform prominence tends to lower them. The same contrast helps explain why the price effects of hybrid forms of prominence are ambiguous. When prominence is partly uniform and partly personalized, the net effect depends on the balance between the change in the fresh–returning demand mix and the front-loading effect induced by personalization.²⁵

8.5. Consumer surplus

Table IV reports the changes in consumer surplus generated by the RS after firms have adjusted their prices. A comparison with Table II shows that the resulting price increase absorbs more than half of the gains from improved matching and reduced search that would arise if prices remained fixed. Under the conservative-beliefs assumption considered here, however, it is not large enough to offset these benefits entirely. In the baseline specification, consumer surplus therefore remains above its random-search level.²⁶

²⁵For example, [Zhong \(2023\)](#) studies an environment in which information directs consumers to a subset of products with high match values, without ranking products within that subset. In their model, more precise information can reduce prices unless it becomes sufficiently precise that consumers stop searching altogether. Similarly, [Zhou \(2022\)](#) shows that the price effect of pre-search information depends on how information changes consumers’ search incentives: prices fall when consumers search less, and may increase when they search more. Another important difference from our setting is that both papers assume entirely idiosyncratic preferences. They therefore abstract from the systematic similarities among consumers and products that recommendation algorithms exploit, and from the resulting composition effect on fresh demand.

²⁶This is not always true across the range of parameter values we consider in the robustness analysis. For example, Section F.1 of the online appendix shows that surplus may decrease when search costs are higher than in the baseline.

m	7	13	19
Collaborative filtering RS	0.0242 (0.0008)	0.0739 (0.0008)	0.0948 (0.0012)
Content-based RS	0.0227 (0.0008)	0.0696 (0.0007)	0.0824 (0.0010)

TABLE IV: Changes in consumer surplus under conservative beliefs in model units.

8.6. Multiple recommendations

We have so far assumed that the platform recommends a single product. The front-loading effect continues to operate, and is in fact amplified, when the platform provides a short ordered list instead. Specifically, suppose that each consumer receives a personalized list of two or three products, ranked by estimated match value, and inspects them in that order. If the consumer has not stopped searching after exhausting the list, the remaining products are inspected in random order.

No. of recommended products m	2			3		
	7	13	19	7	13	19
Collaborative filtering RS	13.18 (0.36)	29.07 (0.55)	44.78 (0.74)	13.26 (0.36)	29.63 (0.55)	46.40 (0.75)
Content-based RS	14.04 (0.32)	32.14 (0.48)	52.24 (0.64)	14.11 (0.32)	32.63 (0.49)	53.53 (0.65)

TABLE V: Percentage changes in equilibrium prices with multiple recommendations.

Table V shows that multiple recommendations raise prices further relative to the single-recommendation case. A longer recommendation list strengthens front-loading by exposing firms to an even more favorably selected pool of consumers. However, the incremental effect is modest: in our calibration, many consumers already buy the first product they inspect, so additional recommendations affect only a limited share of search paths.

9. INFORMATION AND WELFARE

This section has two objectives. The first is to assess the robustness of the main result presented in the previous section. We show that prices continue to increase when we vary the density of the observed ratings matrix and the noise in consumer feedback. The result is also robust to changing the process that generates the data used to train the algorithms, from random data to endogenous data that may exhibit a feedback loop, as discussed in Section 4.

These exercises all vary, directly or indirectly, the amount and quality of information available to the algorithms.²⁷ The second objective of the section is therefore to study comparative

²⁷Section F of the online appendix shows that the price effect is also robust to changes in the search cost, the number of consumers and products, and the number of latent factors, none of which are clean shifts in the algorithm's information. It is also robust to changes in β , which affects the informativeness of ratings but also the price elasticity of demand. Its effect on prices therefore cannot be attributed solely to information.

statics with respect to algorithmic information. We show that more information leads to higher prices and may reduce consumer welfare.

9.1. Varying information levels

We change algorithmic information in three ways. First, we vary the density d of the observed ratings matrix $\tilde{\mathbf{R}}$. In the baseline specification, $d = 1.2\%$. We consider values ranging from 0.6% to 2.4% , in increments of 0.3% .

Second, we vary the precision of consumers' feedback. In the baseline specification, the standard deviation of the reporting noise ϵ_{ij} is equal to 20% of the standard deviation of expected utility across the J products. We let this value range from 0 to 40% , in increments of 2.5% . An alternative way to vary the informativeness of ratings is to consider coarse feedback. Consumers often report evaluations on a Likert scale, such as star ratings, and platforms may also infer coarse signals from observed behavior, for example whether a consumer purchases an inspected product or watches a movie to completion. To capture this possibility, we partition the range of reported utilities $\tilde{u}_{ij}$ into k equal-sized intervals and assume that the algorithm observes only the interval containing the rating, rather than the exact value of $\tilde{u}_{ij}$. We consider values of k between 2 and 10 .

Finally, we consider data generated endogenously, as described in Section 4 and in greater detail in the online appendix. As discussed there, moving from randomly generated data to endogenous data reduces the quality of the information available to the algorithm.

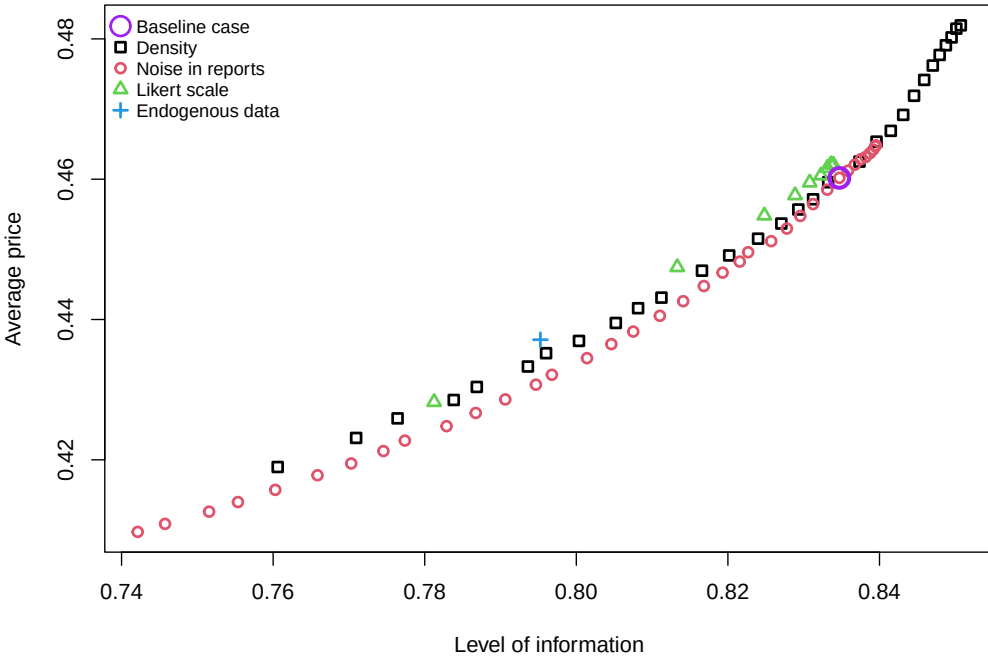


FIGURE 2.—The effect of information on equilibrium prices.

Section G of the online appendix reports detailed results for each exercise. A notable pattern emerges: outcomes are driven primarily by the informativeness of the recommendations rather than by the particular source of variation. We can therefore pool these cases into a single comparative-statics exercise in the precision of algorithmic recommendations. We measure this precision by the average net utility consumers would obtain from the recommended product at zero prices.

9.2. Information and prices

Figure 2 plots the average equilibrium price against the level of information available to the algorithm. Prices rise at an increasing rate with the quantity and quality of information. Since we continue to assume conservative beliefs in this section, this pattern reflects the mechanical effect of greater market segmentation generated by the stronger front-loading of high-value consumers.

Since the market is covered, higher equilibrium prices translate directly into higher firm profits. Firms therefore unambiguously benefit from increases in the information available to the platform.²⁸

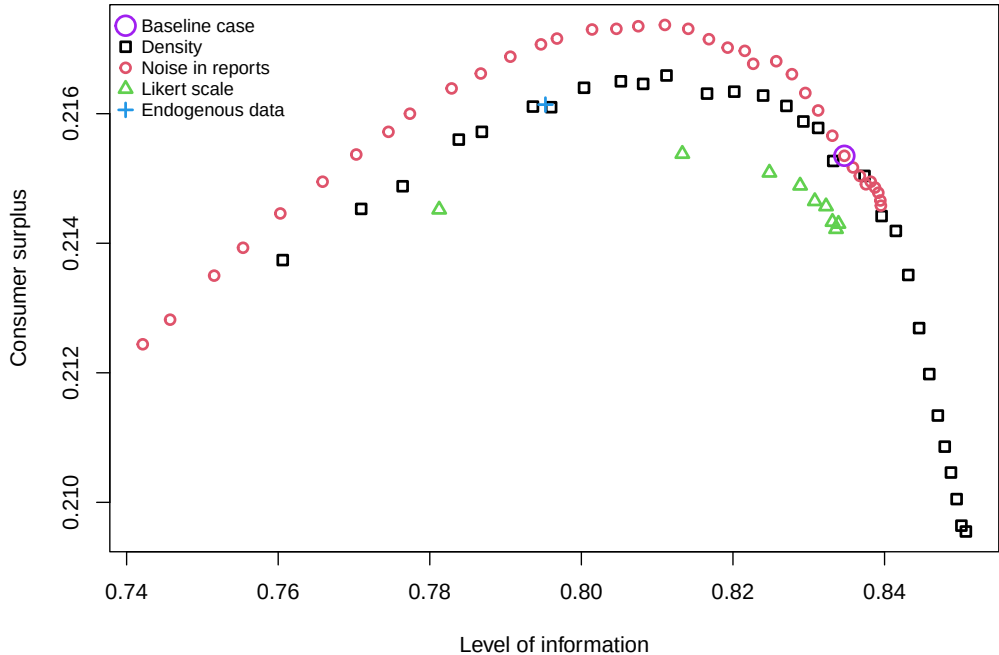


FIGURE 3.—The effect of information on consumer surplus .

²⁸The platform also benefits whenever its revenue increases with transaction prices, as would be the case, for example, under an *ad valorem* fee.

9.3. Information and consumer welfare

More and better information has two opposing effects on consumer welfare. On the one hand, it improves matching and reduces total search costs. On the other hand, it raises equilibrium prices. As shown above, the price effect strengthens as recommendations become more precise. By contrast, the matching gains exhibit diminishing returns: once the algorithm's estimates are already sufficiently accurate, additional information yields only limited further improvements in matching.

Figure 3 shows that, as a result, the welfare effect of information can be non-monotone: the positive effect of improved matching may dominate at low levels of information, while the negative effect of higher prices dominates once recommendations become sufficiently precise. The result is an inverted-U relationship between consumer welfare and information.²⁹

This suggests that consumers may benefit from limits on the amount of personal information available to platforms. Some degree of privacy may therefore be valuable not only in itself, but also because it can preserve the intensity of product-market competition.

10. THE RELIANCE EFFECT

We have so far assumed that consumers hold conservative beliefs about continued search. They use the recommendation to choose the first product to inspect, but evaluate any subsequent search using the same match-value distribution as in the random-search benchmark. This isolates the front-loading effect. Once consumers recognize that recommendations are informative, however, they should also infer that products not recommended to them are, on average, less attractive than a randomly drawn product.

This section studies the consequences of this inference. The inference lowers the expected return from continued search after the recommended product has been inspected. Consumers are therefore willing to accept a lower realized surplus rather than search further, making them more captive to the first firm they visit. This *reliance effect* weakens competitive pressure and induces firms to raise equilibrium prices further.

The magnitude of the reliance effect depends on how strongly consumers revise their beliefs. In the results below, we impose rational expectations: consumers know the recommendation technology and correctly infer the distribution of match values among products that are not recommended. Table VI reports the resulting equilibrium prices for our baseline scenario.

The reliance effect is large. With rational beliefs, the increase in price-cost margins generated by RSs is several times larger than when recommendations operate only through front-loading. As a result, the price increase now more than offsets the direct consumer benefits of recommendations, as shown in Table VII.

²⁹The trade-off identified here, namely, that more information improves matching but also strengthens firms' market power and raises prices, also appears in other settings, such as [Armstrong and Zhou \(2022\)](#) and [Jullien and Pavan \(2019\)](#). Those papers, however, study environments without consumer search, and the mechanisms behind the trade-off differ from ours. The trade-off identified by [Zhou \(2022\)](#), which also arises in a search model, operates through a different channel. In his framework, pre-search information induces consumers to search more, whereas in our baseline model recommendations reduce search. Zhou's mechanism is closer to an extension of our framework, briefly discussed in the conclusion, in which multiple recommendations may increase consumers' search.

m	7	13	19
Collaborative filtering RS	72.68 (3.53)	99.09 (2.98)	104.92 (2.38)
Content-based RS	78.95 (3.10)	113.98 (3.04)	132.73 (2.38)

TABLE VI: Percentage changes in equilibrium prices under rational expectations.

However, these results should be interpreted with some caution. The strength of the reliance effect depends on how accurately consumers understand the informational content of recommendations. The rational-expectations assumption may overstate consumers' ability to assess the informativeness of the RS and, consequently, the extent to which they revise downward the value of continued search. We have therefore also considered intermediate cases in which the search cutoff is adjusted only partially. The resulting price increase is monotone in the extent of this adjustment: it is smallest under conservative beliefs and largest under rational expectations.

Ultimately, the extent to which consumers update their beliefs in response to recommendations is an empirical question. Our results nevertheless highlight a somewhat paradoxical implication: the more consumers trust the platform's recommendations, the less they may ultimately benefit from them.

m	7	13	19
Collaborative filtering RS	-0.1986 (0.0140)	-0.1343 (0.0100)	-0.0942 (0.0074)
Content-based RS	-0.2160 (0.0142)	-0.1748 (0.0107)	-0.1675 (0.0079)

TABLE VII: Changes in consumer surplus under rational expectations in model units.

11. DIRECTED SEARCH

So far, we have worked in a Wolinsky–Anderson–Renault type search environment in which consumers know the distribution of equilibrium prices but do not observe individual prices before inspection. In many online markets, however, prices are readily observable. It is therefore natural to assume that consumers observe prices before undertaking the more detailed inspection required to learn their match values. Search is then directed even without recommendations, because consumers can use prices to determine the order in which products are inspected. In this section, we show that our main results extend to this environment: personalized prominence still raises prices, and the increase is amplified by an additional mechanism, which we call the *price-salience effect*. To focus on this new mechanism, throughout this section, we abstract from the reliance effect.³⁰

³⁰In the directed-search analytical framework, this means that consumers compute the cutoff u^* defined in (14) using the prior distribution $F_{\bar{u}}$ of the systematic match component, rather than the posterior distribution induced by the fact that non-recommended products were not selected by the RS.

11.1. *New assumptions*

The directed-search extension requires several departures from the baseline assumptions.

First, since prices are now observed before inspection, recommendations must be based on estimated surplus rather than on estimated match values alone. We therefore assume that the RS ranks products according to $\hat{u}_{ij} - p_j$. This preserves the role of RSs as providing additional information about match values only, while incorporating the price information consumers already observe.

Second, when prices affect the order of search, firms have an additional incentive to price aggressively. Both with and without an RS, a lower price not only raises the probability of purchase conditional on inspection; it can also move the product earlier in consumers' search order. As emphasized in the directed-search literature, this search-order effect can create discontinuities in demand and jeopardize the existence of a pure-strategy price equilibrium. To smooth demand in the benchmark of unassisted search, following this literature, we assume that consumers observe, before inspection, both the price p_j and an informative signal of the match value. Specifically, we assume that consumers observe the idiosyncratic component of value ε_{ij} , but not the systematic match component $\bar{u}_{ij}$. Thus, in the absence of recommendations, the search order depends on $\varepsilon_{ij} - p_j$.

A similar smoothing is needed with the RS. We therefore allow recommendations to be probabilistic. Product j is recommended to consumer i with a probability ρ_{ij} that depends on the relative value of the expected surplus from product j , $\hat{u}_{ij} - p_j$, and from other products. Specifically, we posit a Boltzmann-type formula with temperature η :

$$\rho_{ij} = \frac{e^{\frac{\hat{u}_{ij} - p_j}{\eta}}}{\sum_{s \in M} e^{\frac{\hat{u}_{is} - p_s}{\eta}}}.$$

The deterministic recommendation rule is obtained in the limit as $\eta \rightarrow 0$. In the simulations, we choose β and η just large enough to make demand sufficiently smooth for a pure-strategy equilibrium to exist. Specifically, we set $\beta = 0.08$ and $\eta = 0.1$.

Finally, in this environment it cannot be taken for granted that consumers follow the recommendations. Consumers observe $\varepsilon_{ij} - p_j$, while the algorithm provides information about $\bar{u}_{ij} - p_j$. Starting the search from the recommended product is optimal only if the algorithm's information is sufficiently valuable relative to the pre-search information consumers already possess. If the idiosyncratic component were too important, consumers would instead start from the product with the highest observed value of $\varepsilon_{ij} - p_j$, and recommendations would have no effect.

We must therefore restrict attention to parameter values for which recommendations are sufficiently informative that consumers prefer to follow them rather than rely on their own information. In other words, holding prices constant, the expected surplus from following the recommendation must exceed that obtained by inspecting first the product with the highest value of $\varepsilon_{ij} - p_j$. This condition holds for all parameter values considered in this section. The results should not, however, be extrapolated to environments in which the shocks ε_{ij} are substantially more dispersed or recommendations are substantially less informative.

11.2. *Demand under directed search*

In the case without RSs, demand can be derived from the analysis in [Choi et al. \(2018\)](#) (see Section H of the online appendix for further details). Their Theorem 1 implies that demand

is the same as if consumers inspected all products and chose the one yielding the highest net surplus, except that the systematic match value $\bar{u}_{ij}$ is replaced by the capped value $\mu_{ij} = \min\{\bar{u}_{ij}, u^*\}$, where the cutoff u^* solves

$$\int_{u^*}^{\infty} (\bar{u} - u^*) dF_{\bar{u}}(\bar{u}) = c_s. \quad (14)$$

Since the idiosyncratic shocks are Gumbel distributed, the probability that consumer i purchases product j is then

$$X_{ij} = \frac{e^{\frac{\mu_{ij} - p_j}{\beta}}}{\sum_{s \in M} e^{\frac{\mu_{is} - p_s}{\beta}}}. \quad (15)$$

To obtain the demand functions with an RS, we proceed as follows. When consumer i inspects product r_i first, Proposition 1 in [Choi et al. \(2018\)](#) implies that the recommended product enters the purchase decision with its full systematic match value, while non-recommended products enter with their capped values. Thus, the probability that consumer i purchases the recommended product r_i is $X_{ir_i} = \frac{e^{(\bar{u}_{ir_i} - p_{r_i})/\beta}}{\Theta}$, where

$$\Theta = e^{\frac{\bar{u}_{ir_i} - p_{r_i}}{\beta}} + \sum_{s \in M \setminus \{r_i\}} e^{\frac{\mu_{is} - p_s}{\beta}},$$

while for any non-recommended product $j \neq r_i$, the purchase probability is $X_{ij} = \frac{e^{(\mu_{ij} - p_j)/\beta}}{\Theta}$. Since recommendations are probabilistic, the unconditional purchase probability is obtained by averaging over possible recommendations. Expected demand is then obtained by summing across consumers.

Relative to the case without recommendations, the capped value μ_{ir} of the recommended product is replaced with the full value $\bar{u}_{ir}$. Since $\bar{u}_{ir} \geq \mu_{ir}$, being recommended raises the probability that the product is purchased. This is the analogue of the front-loading effect in the directed-search environment.

11.3. The price-salience effect

Table [VIII](#) reports the percentage changes in equilibrium prices with and without the RS under these new assumptions. Algorithmic recommendations again raise prices. Because the reliance effect is shut down by construction, the increase must come from other forces.

The first is the familiar front-loading effect. In this directed-search environment, however, front-loading is weaker than in the baseline for two reasons. Firstly, unassisted search is no longer random: consumers already observe $\varepsilon_{ij} - p_j$, so some sorting toward attractive products occurs even without recommendations. Secondly, recommendations are probabilistic, which makes them less informative than in the baseline. Recommendations therefore add less incremental sorting of high-match consumers towards the firms they value most.

But even if the front loading is attenuated, Table [VIII](#) shows that in some cases prices increase more than in the baseline. For example, when $m = 7$, the price increase is more than twice as large as in the baseline. This suggests that an additional force is at work.

m	7	13	19
Collaborative filtering RS	38.85 (1.60)	40.36 (0.94)	31.52 (0.47)
Content-based RS	39.04 (1.59)	40.45 (0.95)	31.58 (0.48)

TABLE VIII: Percentage changes in equilibrium prices under directed search.

This force operates through the role of prices in determining the order of inspection. With or without recommendations, a lower price can move a product earlier in a consumer’s search order. But price is less decisive when recommendations are present. Without the RS, search rankings depend on prices and on the consumer’s idiosyncratic signal, ε_{ij} . With the RS, they depend on prices and on the algorithm’s estimate of the systematic match value. Since consumers follow recommendations only when this algorithmic signal is sufficiently informative relative to their pre-search information, variation in estimated match values plays a relatively large role in determining search order. A given price cut then has a smaller effect on the probability that a product is inspected early. The RS therefore reduces the salience of price in the search process, softens this channel of competition, and puts upward pressure on equilibrium prices. We call this mechanism the *price-salience effect*.

The price-salience effect explains why, in our simulations, recommendations may generate a larger price increase under directed search than under random search. It is important to note, however, that the sign of this effect is ambiguous in general. The reason is that the preceding intuition is incomplete: price salience does not depend solely on the relative informativeness of consumers’ pre-search information and the algorithmic recommendations. The additional incentive to cut price under directed search depends on the probability that a small price reduction changes the order of inspection.

For example, with two products this probability is determined by the distribution of the gap between the largest and second-largest realizations of ε_{ij} without the RS, and by the corresponding gap distribution for the estimated systematic match values, $\hat{u}_{ij}$ with the RS; more precisely, starting from symmetric prices, it is governed by the densities of these two distributions at zero. In our baseline specification, in which the idiosyncratic shocks are Gumbel-distributed and the systematic match values are uniformly distributed, the difference between these densities is closely related to the relative dispersion of ε_{ij} and $\hat{u}_{ij}$, and hence to the relative informativeness of recommendations. For less regular distributions, however, this relationship need not hold, and the sign of the price-salience effect may be ambiguous.

12. SELF PREFERENCING

Finally, we consider the possibility that the platform recommends a product other than the one with the highest estimated match value. This behavior may arise when the platform has an incentive to promote particular products, for example because they generate higher revenues or are supplied by a platform subsidiary rather than an independent supplier. Such self-preferencing has been the subject of several antitrust cases and a growing economic literature,³¹ yet its competitive and welfare effects remain unsettled. Our framework allows us to

³¹Following the seminal contribution of Hagiu and Jullien (2011), more recent papers include Calvano and Jullien (2018), de Cornière and Taylor (2019), Teh and Wright (2022), Bourreau and Gaudin (2022), Peitz and Sobolev (2022), and Bar-Isaac and Shelegia (2023).

revisit this debate while accounting explicitly for how biased recommendations affect consumer search and equilibrium prices.³²

Because products are symmetric *ex ante*, we select the favored product at random. Initially, we treat as exogenous the frequency with which the platform recommends this product to consumers for whom it is not the best estimated match; we refer to this frequency as the *manipulation rate*. Even for a given manipulation rate, however, the platform must decide which consumers to target. We assume that it promotes the favored product to those consumers for whom it is closest to the estimated ideal product. This targeting rule maximizes the probability that the favored product is ultimately purchased while minimizing the potential costs of manipulation.³³

Algorithmic estimates therefore continue to play two roles even under manipulation. They determine which consumers are targeted when the platform manipulates, and which product is genuinely recommended when it does not.

Table IX reports the percentage change in equilibrium prices with a manipulation rate of 20%, relative to the case of genuine recommendations. We report the results for $m = 13$ and for collaborative-filtering RSs; those for other market sizes and content-based systems are similar.

m	7	13	19
Conservative beliefs	−3.27 (0.24)	−5.44 (0.27)	−8.12 (0.32)
Rational expectations	−14.03 (0.72)	−16.98 (0.63)	−18.58 (0.60)

TABLE IX: Percentage changes in equilibrium prices relative to genuine recommendations with a 20% manipulation rate.

Manipulation lowers equilibrium prices under both conservative beliefs and rational expectations. It does so through three channels. First, it weakens the front-loading effect by sorting consumers less accurately toward products they value highly. Second, it makes prominence more uniform, which, as discussed above, tends to intensify price competition. These two channels operate even under conservative beliefs. Under rational expectations, a third channel arises, making the price reduction substantially larger: consumers correctly recognize that manipulated recommendations are less informative and therefore assign a higher value to continued search. They consequently search more, strengthening competitive pressure and inducing firms to lower their prices. Put differently, manipulation attenuates the reliance effect.

³²To our knowledge, [Teh and Wright \(2022\)](#) is the only paper that studies the possibility of manipulating recommendations in a search framework. In their model, however, manipulation occurs only off equilibrium. In equilibrium, the platform recommends to each consumer the product that is best suited to her, and consumers purchase it without further search. Nevertheless, the threat of manipulation induces firms to offer the platform higher transaction fees, which in turn leads to higher equilibrium prices.

³³Misleading recommendations may lead consumers to purchase through alternative channels or to stop using the platform, reducing its revenue or user base. While we do not model these costs explicitly, nor any direct reputational penalty from perceived manipulation, their presence may rationalize our targeting rule.

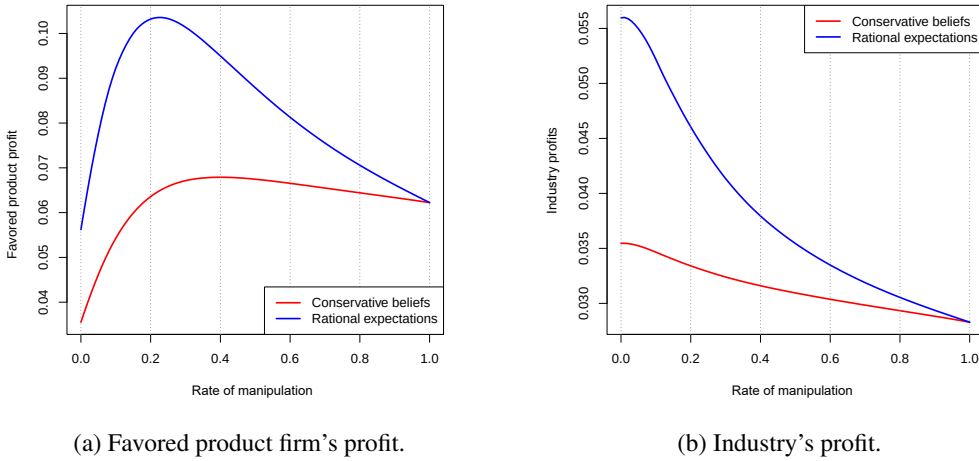


FIGURE 4.—Profits of the favored firm (*Left panel*) and industry profits (*Right panel*) as a function of the manipulation rate.

m	7	13	19
Conservative beliefs	0.0038 (0.0005)	0.0015 (0.0004)	0.0069 (0.0008)
Rational expectations	0.0949 (0.0069)	0.0953 (0.0049)	0.0972 (0.0045)

TABLE X: Changes in consumer surplus relative to genuine recommendations with a 20% manipulation rate.

Table X reports the corresponding changes in consumer surplus. Here, manipulation has two opposing effects. On the one hand, consumers receive less suitable recommendations; on the other, they benefit from lower equilibrium prices. Table X shows that in our simulations, the latter effect dominates, so manipulation increases consumer surplus overall. The gain is modest under conservative beliefs but substantially larger under rational expectations.

The stronger competition induced by self-preferencing may, in turn, limit the profitability of the practice itself. Panel (a) of Figure 4 plots the profits of the firm supplying the favored product as a function of the manipulation rate. Profits are maximized at relatively modest levels of manipulation, especially under rational expectations, and decline as manipulation becomes more extensive.³⁴ Self-preferencing also reduces the profits of competing firms and, as Panel (b) shows, lowers aggregate industry profits. These effects on rivals may be of greater policy concern than the immediate consequences for consumer welfare, since persistently lower profits may deter entry or induce exit in the longer run. In this sense, self-preferencing may be better understood as a potentially exclusionary practice than as an exploitative one.

³⁴If the platform's revenue is proportional to transaction prices, as under ad valorem fees, the manipulation rate that maximizes the joint profits of the platform and the firm supplying the favored product is even lower.

13. CONCLUSIONS

In this paper, we have studied how algorithmic recommendations affect product-market competition and consumer welfare. Our analysis identifies two distinct collective-action problems among consumers. First, each consumer may individually benefit from providing the platform with more accurate feedback, because doing so improves the quality of the recommendations she receives. If all consumers provide more informative feedback, however, the resulting increase in the platform's information raises equilibrium prices and may ultimately reduce consumer welfare. Thus, behavior that is individually optimal may leave consumers collectively worse off. This creates scope for coordination among consumers or for forms of regulation that limit the information available to platforms. Privacy regulation, for example, may benefit consumers not only by protecting their private sphere, but also by intensifying competition.

A similar collective-action problem arises in consumers' interpretation of recommendations. Each consumer has an individual incentive to strive to understand their informational content as accurately as possible, since this allows her to assess the value of continued search and make a better-informed stopping decision. When all consumers do so, however, the resulting reliance effect weakens competition and leads to higher equilibrium prices. Paradoxically, consumers may be harmed by recommendations precisely when they trust them most.

On the other hand, an accurate understanding of the informational content of recommendations may collectively protect consumers against the platform's incentive to manipulate them. When the platform engages in manipulation, rational consumers recognize that recommendations are less informative and rely on them less. This induces more search, intensifies price competition, and limits the profitability of manipulation.

We conclude by briefly mentioning possible directions for future research. Although we have conducted a broad set of robustness checks, several important extensions remain. First, we have analyzed multiple recommendations only under conservative beliefs and unobservable prices. With rational expectations, an ordered list conveys information about both recommended and non-recommended products. The expected value of searching among non-recommended products falls, as in our analysis of a single recommendation, but the value of inspecting additional recommended products may rise. The resulting effect on the continuation value of search is therefore ambiguous. Multiple recommendations also make directed search more complex, because consumers must decide whether to follow the platform's ranking or reorder products using their own pre-search information, including observed prices.

Second, it would be interesting to introduce competition among platforms. Platform competition may affect incentives to collect data, improve recommendation technology so as to acquire a competitive advantage, or manipulate rankings.

Third, a more dynamic framework could endogenize product entry and exit, quality choice, and R&D investment. These extensions are especially relevant to self-preferencing, since its short-run effect of lowering prices may coexist with longer-run exclusionary effects.

Finally, an intriguing possibility is that algorithmic recommendations may shape consumer preferences rather than merely predict them. As recognized at least since [Knight \(1923\)](#), tastes may be endogenous. Inspecting a product—for example, beginning to watch a movie or listen to a piece of music—may not only reveal its match value, but also influence the consumer's preferences. If platforms can affect this process through their recommendations, RSs become more than informational devices: they also become instruments for shaping tastes. This raises new questions about platform incentives and consumer welfare.

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